

The SC-at-1 trivial reduction of Converse Diagnostic 4

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Abstract

We give a trivial proof closing **378/514** $\approx$ **73.5%** of the empirically tested converse Diagnostic 4 cases. For $T \in \text{SYT}(\lambda)$ with $\lambda = (\hat{\lambda}, 1^q)$ and $\tau(\hat{\lambda}) = 0$, if the cell containing 2 in T is directly below the cell containing 1 (an SC pair at index 1), then $S_1 v_T = 0$ in the Hoefsmit basis of V^λ . Hence $R'_n v_T = 0$, and thus $\Omega v_T = 0$ and $p_T(q) = 0$. Combined with yesterday's universal SC necessity proof, this fully closes Diagnostic 4 (forward and converse) for the SC-at-1 stratum. The conjecture's remaining open core is the SR-at-1 stratum: 136 zero diagonals out of 514, all of which Ω - or Ω^* -kernel-vanish empirically, but for which the chain-product collapses at deeper combinatorial events.

1 Setting

Fix $\lambda = (\hat{\lambda}, 1^q)$ with $\ell(\hat{\lambda}) \geq 2$, every row $\hat{\lambda}_r \geq 2$, and $\tau(\hat{\lambda}) := \max(0, q + 2\ell(\hat{\lambda}) - 1 - |\hat{\lambda}|) = 0$. Let $n = |\lambda|$, $\ell = \ell(\lambda)$, and let V^λ denote the irreducible $\mathcal{H}_q(\mathfrak{S}_n)$ -module realised in the Hoefsmit (seminormal) basis indexed by $\text{SYT}(\lambda)$. We write T_i for the Hecke generators with $(T_i - q)(T_i + 1) = 0$ and put $S_i := T_i + 1$.

Following the conventions of yesterday's note `2026-05-18-converse-diagnostic-4.tex`, define

$$\begin{aligned} R'_k &:= S_{k-1} S_{k-2} \cdots S_2 S_1, \\ L_k &:= S_1 S_2 \cdots S_{k-2} S_{k-1}, \\ \Omega &:= R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \\ \Omega^* &:= L_n L_{n-1} \cdots L_{\ell+1}, \end{aligned}$$

and write $v_T \in V^\lambda$ for the seminormal basis vector indexed by T . The diagonal probe is

$$p_T(q) := v_T \Omega v_T \in \mathbb{Q}(q),$$

where $\cdots$ pairs a basis vector with the coefficient of v_T in the expansion.

Definition 1. An *SC pair* of T at index i is a pair $(i, i+1)$ such that the cells containing i and $i+1$ in T share a column. An *SR pair* is the row-analogue. We say T is *SC-at-1* if the cell containing 2 is $(2, 1)$ (so 2 is directly below 1); equivalently, the SC pair $(1, 2)$ is present. Otherwise (since 2 is either at $(1, 2)$ or $(2, 1)$ in any SYT) we say T is *SR-at-1*.

2 The trivial step

Lemma 2. *For $T \in \text{SYT}(\lambda)$, if T is SC-at-1 then $S_1 v_T = 0$.*

Proof. By Hoefsmit's seminormal formula, if entries $i, i+1$ of T share a column, then $T_i v_T = -v_T$. Applying this at $i = 1$ for an SC-at-1 T gives $T_1 v_T = -v_T$, whence $S_1 v_T = (T_1 + 1)v_T = 0$. $\square$

Lemma 3. *For any vector $w \in V^\lambda$ with $S_1 w = 0$ we have $R'_n w = 0$. In particular, if T is SC-at-1 then $R'_n v_T = 0$.*

Proof. By definition $R'_n = S_{n-1} S_{n-2} \cdots S_2 S_1$, whose rightmost factor is S_1 . Hence $R'_n w = S_{n-1} \cdots S_2 (S_1 w) = S_{n-1} \cdots S_2 \cdot 0 = 0$. Combined with Lemma 2, $S_1 v_T = 0$, so $R'_n v_T = 0$. $\square$

Theorem 4 (SC-at-1 trivial reduction). *Let λ be a $\tau = 0$ shape as above. For every $T \in \text{SYT}(\lambda)$ with T SC-at-1,*

$$\Omega v_T = 0 \quad \text{and consequently} \quad p_T(q) = 0.$$

Proof. By Lemma 3, $R'_n v_T = 0$. Then

$$\Omega v_T = R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} (R'_n v_T) = R'_{\ell+1} \cdots R'_{n-1} \cdot 0 = 0,$$

so $\Omega v_T = 0$ and the diagonal entry $p_T(q) = v_T \Omega v_T = 0$. $\square$

Remark 5. Theorem 4 is striking only against the backdrop of the forward direction. Yesterday's note showed $p_T = 0 \Rightarrow T$ has some SC pair, but the SC pair could be at any index. Theorem 4 says that if it is at the *specific* index 1, then converse Diagnostic 4 closes by a one-line algebraic identity: the Hecke factor S_1 at the right end of R'_n already kills v_T . No further argument is needed.

3 Computational check

In a sample of 12 shapes

$$\lambda \in \{(3, 2), (3, 2, 1), (2, 2, 1), (3, 2, 2), (3, 2, 1, 1), (3, 3, 2), (3, 2, 2, 1), (3, 3, 2, 1), (4, 3), (4, 3, 1), (4, 3, 2), (3, 3, 1, 1, 1)\}$$

covering every previously-tested converse target (cf. `2026-05-18-converse-probe/dichotomy_verifier.py`), we symbolically verified $S_1 v_T = 0$ for every T with 2 in cell (2,1) in the Hoefsmit basis. The count was 378 out of 378 such T (i.e. no failures). Cross-checked against the stratification trace `2026-05-19-stratification/stratify.py`, all 378 SC-at-1 T produced the collapse step $(k_R, i_R) = (n, 1)$, meaning the very first factor S_1 of R'_n vanishes the basis vector.

This is a confirmation of Lemma 2 on actual matrices, not a separate piece of evidence: Hoefsmit's formula is exact and the identity $S_1 v_T = 0$ is a direct calculation. We report it because matching the predicted collapse step at the symbolic level is the cleanest external check.

4 Consequence: the residual open problem

Combining Theorem 4 with yesterday's universal SC necessity (every T with $p_T = 0$ has at least one SC pair) gives:

Corollary 6. *For λ of the form above, the set of zero diagonals decomposes as*

$$\{T \in \text{SYT}(\lambda) : p_T = 0\} = \{T : T \text{ is SC-at-1}\} \sqcup \{T : T \text{ is SR-at-1 and } p_T = 0\}.$$

The first set is closed under converse Diagnostic 4 by Theorem 4: every such T satisfies $\Omega v_T = 0$.

The full converse conjecture therefore reduces to:

For every $T \in \text{SYT}(\lambda)$ with T SR-at-1 (i.e. $2 \in (1, 2)$) and $p_T(q) = 0$, either $\Omega v_T = 0$ or $\Omega^* v_T = 0$.

In our 12-shape corpus the SC-at-1 stratum contributes $378/514 \approx 73.5\%$ of all zero diagonals. The SR-at-1 residual contains 136 zero diagonals.

5 Empirical scaffolding for the residual

Although the SR-at-1 converse is not closed in this note, the stratification trace produces tight constraints. We summarise what the data force.

Right-collapse normal form. For every SR-at-1 zero diagonal in the corpus, the collapse step (k_R, i_R) of the right chain Ω , if finite, has the form $(k_R, 1)$: *the leftmost factor S_1 of R'_{k_R} vanishes the running vector*. Equivalently, the partial product $R'_{k_R+1} R'_{k_R+2} \cdots R'_n v_T$ already lies in $\ker S_1$, i.e. the eigenspace where T_1 acts by -1 :

$$\text{span}\{v_S : S \in \text{SYT}(\lambda), S \text{ is SC-at-1}\}.$$

Left-collapse normal form. For every SR-at-1 zero diagonal whose Ω^* -collapse is finite, $(k_L, i_L) = (k_L, k_L - 1)$: *the rightmost factor S_{k_L-1} of L_{k_L} vanishes the running vector*. Equivalently the partial product $L_{k_L-1} \cdots L_{\ell+1} v_T$ lies in $\ker S_{k_L-1}$.

Ω -survivors. 12 of the 136 SR-at-1 zeros have $\Omega v_T \neq 0$ (and are killed only by Ω^*). Every one of them has either

- (i) T contains 1, 2, 3 in row 1 (11 of 12),
- (ii) T has 1, 2 in row 1 and 3, 4 in row 2 at cells $(2, 1), (2, 2)$ (1 of 12).

In both cases T has at least *two* SR pairs at small indices, preventing the chain from steering into $\ker S_1$ during Ω .

This phenomenology suggests the residual conjecture splits into a *double-SR-prefix* sub-case (handled by Ω^*) and a *single-SR-prefix* sub-case (handled by Ω).

Conjecture 7 (Stratified converse Diagnostic 4). *Let $T \in \text{SYT}(\lambda)$ with T SR-at-1 and $p_T(q) = 0$.*

- (i) *If T does not contain 1, 2, 3 in row 1 and 3, 4 do not both lie in row 2 at $(2, 1), (2, 2)$, then $\Omega v_T = 0$.*
- (ii) *Otherwise $\Omega^* v_T = 0$.*

Conjecture 7 holds for all 136 SR-at-1 zero diagonals in the corpus. Its proof remains open.

6 Where this leaves Route A

The Hecke-algebraic stratification (Route A) of the prove session's plan is now *partially closed*:

- The SC-at-1 stratum closes by Theorem 4.
- The SR-at-1 stratum reduces to Conjecture 7, whose two halves each ask for a different structural identity in the chain product.

The two halves are not symmetric: SR-at-1 T's break the left/right duality that the SC-at-1 trivial proof exploits. This matches the 2026-05-19 refutation of universal SR necessity (2026-05-19-sr-conjecture-ref).

Sharper next step. In the right-collapse normal form, the chain product $R'_{k_R+1} \cdots R'_n v_T$ projects v_T into the SC-at-1 eigenspace. Identifying the smallest k at which this projection becomes total — and matching it to combinatorial data on T — is the natural next problem. This is a question about how the “letter-shift” operators R'_k act on the position of 2 in the support of the running vector.

Acknowledgements

This note builds on yesterday’s universal SC necessity (2026-05-18-converse-diagnostic-4.tex), the two-sided kernel dichotomy (2026-05-18-d-class-row-zero.tex), and the SR-necessity refutation (2026-05-19-sr-conjecture-refuted.tex). The Hoefsmit basis machinery is from 2026-05-08-rank-Pi/Stratification data from 2026-05-19-stratification/stratify.py.